Boundary Element Simulation of Chamfered Pelvic Floor Implants on HPC/NVQLink Hybrid Infrastructure: A Continued-Fraction Manifold Repair Framework

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Abstract: We present a finite mathematical framework for the design and biomechanical validation of chamfered pelvic floor reconstruction implants. Implant boundary geometry is generated with a parametric chamfer profile and repaired to a closed two-manifold using discrete Euler-characteristic and Gauss–Bonnet consistency checks. A continued-fraction convergent expansion of an irrational rotation number parameterizes a quasi-periodic curvature blending field across the chamfer-to-body transition, analogous to KAM-type irrational winding, eliminating periodic stress risers. The resulting closed-manifold surface is discretized into constant boundary panels and solved via the Boundary Element Method (BEM) using the Kelvin fundamental solution under cyclic Valsalva pressure loading. The dense BEM influence system is distributed across an HPC cluster using SLURM-scheduled MPI domain decomposition, with strong-scaling behaviour projected via Amdahl's law, and further accelerated through a simulated NVQLink hybrid GPU–QPU interconnect implementing an HHL-class quantum linear-solver preconditioner.

1. Chamfer Geometry and Manifold Repair

The implant boundary is offset inward by the chamfer width w_c and projected to depth along the bevel angle θ_c, giving the chamfer depth:

$$d_c = w_c \cdot tan(\theta_c)$$

Mesh repair enforces manifold closure by fan-triangulating detected boundary loops and verifying the discrete Euler characteristic $\chi = V - E + F$ against the Gauss–Bonnet identity $(1/2\pi)\blacksquare_M K\, dA + \blacksquare_{\partial M} \kappa_g\, ds = \chi$, confirming genus-0 closure of the repaired implant surface.

2. Continued-Fraction Quasi-Periodic Curvature Blending

The regular continued fraction of a rotation number x (e.g. the golden ratio ϕ, chosen for its maximally irrational Diophantine properties) is expanded as:

$$x = a_0 + 1/(a_1 + 1/(a_2 + 1/(a_3 + \ldots)))$$

with convergents p_k/q_k satisfying $p_k = a_k p_{k-1} + p_{k-2}$, $q_k = a_k q_{k-1} + q_{k-2}$. Superposing convergent harmonics along the boundary arclength s produces the curvature blending field $\kappa(s) = \Sigma_k \cos(2\pi q_k s + p_k) / (|q_k| + 1)$, which is non-resonant by construction and evenly distributes curvature transition across the chamfer-to-body junction.

3. Boundary Element Formulation

Implant biomechanics are evaluated via the Somigliana boundary integral identity for 3D elastostatics on the closed surface Γ:

$$c_{ij}(x)\, u_j(x) + \int_\Gamma T_{ij}(x,y)\, u_j(y)\, d\Gamma = \int_\Gamma U_{ij}(x,y)\, t_j(y)\, d\Gamma$$

with the Kelvin fundamental solution $U_{ij}(x,y) = [1/(16\pi\mu(1-\nu)r)] \cdot [(3-4\nu)\delta_{ij} + r\blacksquare_i r\blacksquare_j]$, discretized into constant boundary panels and collocated to assemble the dense linear system Hu = Gt under cyclic Valsalva intra-abdominal pressure loading (3.5–25 kPa).

4. HPC Domain Decomposition and Amdahl Scaling

The $O(N^3)$ dense BEM solve is distributed across P MPI ranks via SLURM-scheduled domain decomposition. Strong-scaling speedup and parallel efficiency follow Amdahl's law:

$$S(P) = 1 / [(1-f) + f/P], \quad E(P) = S(P)/P$$

where f is the parallelizable fraction of the assembly/solve workload. Communication overhead from boundary-panel halo exchange is modeled to scale with $\sqrt{P}$, bounding the achievable efficiency at large rank counts.

5. NVQLink Hybrid Quantum-Classical Acceleration

A simulated NVQLink interconnect couples HPC GPU nodes to a QPU co-processor for preconditioning the ill-conditioned Kelvin-kernel subspace. HHL-class quantum linear solvers achieve complexity:

$$T_{quantum} \sim O(\log(N) \cdot \kappa^2 / \varepsilon) \text{ vs. } T_{classical} \sim O(N^3)$$

where κ is the system condition number and ε the target solution tolerance. The resulting hybrid pipeline — QPU-assisted deflation followed by GPU-side iterative refinement across the NVQLink fabric — is projected to reduce wall-clock time for well-conditioned implant subspaces relative to the classical direct solve.

6. Conclusion

Combining chamfer-aware manifold repair, continued-fraction quasi-periodic curvature blending, boundary element biomechanical simulation, and HPC/NVQLink hybrid acceleration provides a computationally tractable, mathematically grounded pipeline for the design and clinical validation of next-generation pelvic floor reconstruction implants.